

On the Geometric Constraint Between Angular Deviations in Semiprime Factorization

1. Purpose of this note

The purpose of this note is to clarify the robustness of the geometric relation observed between the angular deviations

Δ_p and Δ_q , derived from the tangential transport of a semiprime $n = pq$ on its dyadic shell.

In particular, we address:

1. Which hypotheses are **invalidated** by robustness tests.
2. Which structural properties are **preserved** across dataset size and generation variants.
3. Whether **permutation** induces a genuine **change of regime**, and if so, in what precise sense.

This note is not intended to introduce new observables, but to stabilize the interpretation of those already defined.

2. Summary of robustness tests performed

The following tests were conducted:

- **Dataset size scaling:** comparison between $N = 20\,000$ and $N = 500\,000$.
- **Generation variants:** balanced semiprimes with different MSB constraints (top-1 / top-2), when available.
- **Permutation null model:** random reassignment of (p, q) across fixed n , destroying the intrinsic factor coupling while preserving marginal distributions.

For each dataset, the observable

$$S = \Delta_p + \Delta_q$$

was analyzed, together with its dispersion and with the correlation between Δ_p and Δ_q .

3. What is *not* conserved

The robustness tests conclusively show that:

- There is **no scalar conservation law** of the form

$$\Delta_p + \Delta_q = \text{constant (e.g. } \pi \text{)}.$$

- Apparent proximity of $\mathbb{E}[S]$ to π at small sample size ($N \sim 20k$) is a **finite-sample artifact**.

- At large scale ($N = 500k$), the mean and variance of S stabilize to values **incompatible with any constant-sum hypothesis**.

Hence, any interpretation in terms of a conserved scalar quantity must be rejected.

4. What is robustly observed

Despite the absence of scalar conservation, all large-scale datasets share the following properties:

1. **Reduced admissible support**
The distribution of S in real data is significantly narrower than in the permutation null model.
2. **Stable regime at large scale**
At $N = 500k$, the shape of the distribution (mean, variance, quantiles) is stable across dataset variants.
3. **Non-trivial coupling**
The correlation between Δ_p and Δ_q is consistently positive and significantly different from the permuted case.

These properties indicate the presence of a **geometric constraint**, not a conservation law.

5. The role of permutation: is there a “law”?

This leads to the central question:

Is there a “law of permutation” that induces a change of regime?

The answer is **yes, but not in the sense of a new invariant**.

5.1 What permutation does

Permutation destroys the **factorial coupling** between p , q , and n , while preserving:

- the marginal distributions of Δ_p and Δ_q ,
- the global dyadic scale,
- the angular embedding framework.

As a result:

- the joint distribution (Δ_p, Δ_q) becomes effectively **factorized**,
- the admissible domain expands,
- dispersion increases sharply.

5.2 What permutation does *not* do

Permutation does **not** introduce a new structure or a competing law.

It simply removes the geometric constraint imposed by the true factorization.

5.3 Interpretation

Permutation is therefore best understood as a **regime-destroying operation**, not a regime-generating one.

The “law” revealed by permutation is negative:

When the intrinsic coupling between p and q is broken, the constrained geometric regime disappears.

Thus, permutation acts as a **null operator** that restores the unconstrained phase space.

6. Reformulation of the core result

The correct structural statement is:

The factorization $n = pq$ induces a **geometric admissibility constraint** on the pair (Δ_p, Δ_q) .

This constraint is stable at large scale, disappears under permutation, and cannot be reduced to a scalar conservation law.

The information is therefore not carried by S , but by the **geometry of the admissible domain** in the (Δ_p, Δ_q) plane.

7. Consequences for the main document

This robustness analysis justifies the following revisions and reinforcements:

- Abandon any claim of scalar conservation.
- Emphasize **transported geometric structure** rather than invariant quantities.
- Treat permutation explicitly as a **null model** demonstrating the necessity of the factor coupling.
- Focus subsequent analysis on **boundaries, lobes, and admissible regions**, not on projections.

8. Outlook

The robustness tests close the statistical question and open a geometric one:

What is the minimal geometric object that characterizes the admissible domain?

This naturally motivates a functional or variational formulation, where the observed domain is reconstructed as the minimal-complexity structure compatible with empirical data.